Supplemental Online Content

Wang Z, Wang X, Wang Z, et al. Comparison of hepatotoxicity associated with new BCR-ABL tyrosine kinase inhibitors vs imatinib among patients with chronic myeloid leukemia: a systematic review and meta-analysis. *JAMA Netw Open.* 2021;4(7):e2120165. doi:10.1001/jamanetworkopen.2021.20165

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This supplemental material has been provided by the authors to give readers additional information about their work.

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eTable 1: Search strategy (January 2000 to April 2020)

Database	Drug	Search Strategy	Number of hits
PubMed/ Embase	Bosutinib	Search ((((bosutinib[MeSH Major Topic]) OR SKI606) OR SKI-606)) AND ((((((

		Leukemia) OR Philadelphia-Positive Myeloid Leukemias) OR Granulocytic Leukemia, Chronic) OR Chronic Granulocytic Leukemia) OR Chronic Granulocytic Leukemias) OR Granulocytic Leukemias, Chronic) OR Leukemia, Chronic Granulocytic) OR Leukemias, Chronic Granulocytic) Filters: Clinical Trial	
	Dasatinib	Search (((((((dasatinib[MeSH Major Topic]) OR Sprycel) OR BMS 354825) OR 354825, BMS) OR BMS354825) OR BMS-354825)) AND ((((((

		Granulocytic Leukemia, Chronic) OR Chronic Granulocytic Leukemia) OR Chronic Granulocytic Leukemias) OR Granulocytic Leukemias, Chronic) OR Leukemia, Chronic Granulocytic) OR Leukemias, Chronic Granulocytic) Filters: Clinical Trial	
	Nilotinib	Search ((((nilotinib[MeSH Major Topic]) OR Tasigna) OR AMN107)) AND ((((((

		Leukemia) OR Chronic Granulocytic Leukemias) OR Granulocytic Leukemias, Chronic) OR Leukemia, Chronic Granulocytic) OR Leukemias, Chronic Granulocytic) Filters: Clinical Trial	
	Ponatinib	Search ((((((ponatinib[MeSH Major Topic]) OR ponatinib hydrochloride) OR AP24534) OR AP-24534) OR AP 24534)) AND ((((((

		Leukemias, Chronic) OR Leukemia, Chronic Granulocytic) OR Leukemias, Chronic Granulocytic) Filters: Clinical Trial	
	Imatinib	Search ((((((((((((((((Imatinib Mesylate[MeSH Major Topic]) OR Mesylate, Imatinib) OR Imatinib Methanesulfonate) OR Methanesulfonate, Imatinib) OR STI571) OR STI-571) OR STI 571) OR Gleevec) OR Glivec) OR ST 1571) OR ST1571) OR CGP 57148) OR CGP57148B) OR CGP-57148) OR CGP57148) OR Imatinib)) AND ((((((((((((((((

		Leukemia) OR Philadelphia-Positive Myeloid Leukemias) OR Granulocytic Leukemia, Chronic) OR Chronic Granulocytic Leukemia) OR Chronic Granulocytic Leukemias) OR Granulocytic Leukemias, Chronic) OR Leukemia, Chronic Granulocytic) OR Leukemias, Chronic Granulocytic) Filters: Clinical Trial	
Cochrane Library	Bosutinib	#1 (Bosutinib):ti,ab,kw OR (SKI606):ti,ab,kw OR (SKI-606):ti,ab,kw (Word variations have been searched) #2 (Leukemia, Myelogenous, Chronic, BCR-ABL Positive):ti,ab,kw OR (Philadelphia-Positive Myeloid Leukemia):ti,ab,kw OR (Myeloid Leukemias, Chronic):ti,ab,kw OR (Leukemia, Myelogenous, Ph1 Positive):ti,ab,kw OR (Leukemias, Chronic Granulocytic):ti,ab,kw (Word variations have been searched) #3 (Leukemia, Myelogenous, Chronic, BCR-ABL Positive):ti,ab,kw OR (Myelogenous Leukemias, Ph1 Positive):ti,ab,kw OR (Leukemia, Myeloid, Philadelphia Positive):ti,ab,kw OR (Myelocytic Leukemias, Chronic):ti,ab,kw OR (Granulocytic Leukemia, Chronic):ti,ab,kw (Word variations have been searched) #4 (Chronic Myelogenous Leukemia):ti,ab,kw OR (Leukemias, Ph1 Positive Myelogenous):ti,ab,kw OR (Myelogenous Leukemias, Chronic):ti,ab,kw OR (Chronic Granulocytic Leukemia):ti,ab,kw OR (Leukemia, Myelogenous, Ph1 Positive):ti,ab,kw (Word variations have been searched) #5 (Leukemia, Philadelphia-Positive Myeloid):ti,ab,kw OR (Ph1 Positive Myeloid Leukemia):ti,ab,kw OR (Myeloid Leukemia, Chronic):ti,ab,kw OR (Leukemia, Myeloid, Ph1 Positive):ti,ab,kw OR (Leukemia, Granulocytic, Chronic):ti,ab,kw (Word variations have been searched) #6 (Leukemia, Ph1 Positive Myeloid):ti,ab,kw OR (Chronic Myelocytic Leukemia):ti,ab,kw OR (Myeloid Leukemias, Ph1 Positive):ti,ab,kw OR (Granulocytic Leukemias, Chronic):ti,ab,kw OR (Myeloid Leukemia, Philadelphia Positive):ti,ab,kw (Word variations have been searched) #7 #2 or #3 or #4 or #5 or #6 #8 #1 and #7	82
	Dasatinib	#1 (dasatinib):ti,ab,kw OR (Sprycel):ti,ab,kw OR (BMS-354825):ti,ab,kw OR (BMS354825):ti,ab,kw OR (354825, BMS):ti,ab,kw (Word variations have been searched) #2 (Leukemia, Myelogenous, Chronic, BCR-ABL Positive):ti,ab,kw OR (Philadelphia-Positive Myeloid	244

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		Leukemia):ti,ab,kw OR (Myeloid Leukemias, Chronic):ti,ab,kw OR (Leukemia, Myelogenous, Ph1 Positive):ti,ab,kw OR (Leukemias, Chronic Granulocytic):ti,ab,kw (Word variations have been searched) #3 (Leukemia, Myelogenous, Chronic, BCR-ABL Positive):ti,ab,kw OR (Myelogenous Leukemias, Ph1 Positive):ti,ab,kw OR (Leukemia, Myeloid, Philadelphia Positive):ti,ab,kw OR (Myelocytic Leukemias, Chronic):ti,ab,kw OR (Granulocytic Leukemia, Chronic):ti,ab,kw (Word variations have been searched) #4 (Chronic Myelogenous Leukemia):ti,ab,kw OR (Leukemias, Ph1 Positive Myelogenous):ti,ab,kw OR (Myelogenous Leukemias, Chronic):ti,ab,kw OR (Chronic Granulocytic Leukemia):ti,ab,kw OR (Leukemia, Myelogenous, Ph1 Positive):ti,ab,kw (Word variations have been searched) #5 (Leukemia, Philadelphia-Positive Myeloid):ti,ab,kw OR (Ph1 Positive Myeloid Leukemia):ti,ab,kw OR (Myeloid Leukemia, Chronic):ti,ab,kw OR (Leukemia, Myeloid, Ph1 Positive):ti,ab,kw OR (Leukemia, Granulocytic, Chronic):ti,ab,kw (Word variations have been searched) #6 (Leukemia, Ph1 Positive Myeloid):ti,ab,kw OR (Chronic Myelocytic Leukemia):ti,ab,kw OR (Myeloid Leukemias, Ph1 Positive):ti,ab,kw OR (Granulocytic Leukemias, Chronic):ti,ab,kw OR (Myeloid Leukemia, Philadelphia Positive):ti,ab,kw (Word variations have been searched) #7 #2 or #3 or #4 or #5 or #6 #8 #1 and #7	
	Nilotinib	#1 (nilotinib):ti,ab,kw OR (Tasigna):ti,ab,kw OR (AMN107):ti,ab,kw (Word variations have been searched) #2 (Leukemia, Myelogenous, Chronic, BCR-ABL Positive):ti,ab,kw OR (Philadelphia-Positive Myeloid Leukemia):ti,ab,kw OR (Myeloid Leukemias, Chronic):ti,ab,kw OR (Leukemia, Myelogenous, Ph1 Positive):ti,ab,kw OR (Leukemias, Chronic Granulocytic):ti,ab,kw (Word variations have been searched) #3 (Leukemia, Myelogenous, Chronic, BCR-ABL Positive):ti,ab,kw OR (Myelogenous Leukemias, Ph1 Positive):ti,ab,kw OR (Leukemia, Myeloid, Philadelphia Positive):ti,ab,kw OR (Myelocytic Leukemias, Chronic):ti,ab,kw OR (Granulocytic Leukemia, Chronic):ti,ab,kw (Word variations have been searched)	265

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		#4 (Chronic Myelogenous Leukemia):ti,ab,kw OR (Leukemias, Ph1 Positive Myelogenous):ti,ab,kw OR (Myelogenous Leukemias, Chronic):ti,ab,kw OR (Chronic Granulocytic Leukemia):ti,ab,kw OR (Leukemia, Myelogenous, Ph1 Positive):ti,ab,kw (Word variations have been searched) #5 (Leukemia, Philadelphia-Positive Myeloid):ti,ab,kw OR (Ph1 Positive Myeloid Leukemia):ti,ab,kw OR (Myeloid Leukemia, Chronic):ti,ab,kw OR (Leukemia, Myeloid, Ph1 Positive):ti,ab,kw OR (Leukemia, Granulocytic, Chronic):ti,ab,kw (Word variations have been searched) #6 (Leukemia, Ph1 Positive Myeloid):ti,ab,kw OR (Chronic Myelocytic Leukemia):ti,ab,kw OR (Myeloid Leukemias, Ph1 Positive):ti,ab,kw OR (Granulocytic Leukemias, Chronic):ti,ab,kw OR (Myeloid Leukemia, Philadelphia Positive):ti,ab,kw (Word variations have been searched) #7 #2 or #3 or #4 or #5 or #6 #8 #1 and #7	
	Ponatinib	#1 (Ponatinib):ti,ab,kw OR (ponatinib hydrochloride):ti,ab,kw OR (AP24534):ti,ab,kw OR (AP-24534):ti,ab,kw OR (Iclusig):ti,ab,kw (Word variations have been searched) #2 (Leukemia, Myelogenous, Chronic, BCR-ABL Positive):ti,ab,kw OR (Philadelphia-Positive Myeloid Leukemia):ti,ab,kw OR (Myeloid Leukemias, Chronic):ti,ab,kw OR (Leukemia, Myelogenous, Ph1 Positive):ti,ab,kw OR (Leukemias, Chronic Granulocytic):ti,ab,kw (Word variations have been searched) #3 (Leukemia, Myelogenous, Chronic, BCR-ABL Positive):ti,ab,kw OR (Myelogenous Leukemias, Ph1 Positive):ti,ab,kw OR (Leukemia, Myeloid, Philadelphia Positive):ti,ab,kw OR (Myelocytic Leukemias, Chronic):ti,ab,kw OR (Granulocytic Leukemia, Chronic):ti,ab,kw (Word variations have been searched) #4 (Chronic Myelogenous Leukemia):ti,ab,kw OR (Leukemias, Ph1 Positive Myelogenous):ti,ab,kw OR (Myelogenous Leukemias, Chronic):ti,ab,kw OR (Chronic Granulocytic Leukemia):ti,ab,kw OR (Leukemia, Myelogenous, Ph1 Positive):ti,ab,kw (Word variations have been searched) #5 (Leukemia, Philadelphia-Positive Myeloid):ti,ab,kw OR (Ph1 Positive Myeloid Leukemia):ti,ab,kw OR (Myeloid Leukemia, Chronic):ti,ab,kw OR (Leukemia, Myeloid, Ph1 Positive):ti,ab,kw	39

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		OR (Leukemia, Granulocytic, Chronic):ti,ab,kw (Word variations have been searched) #6 (Leukemia, Ph1 Positive Myeloid):ti,ab,kw OR (Chronic Myelocytic Leukemia):ti,ab,kw OR (Myeloid Leukemias, Ph1 Positive):ti,ab,kw OR (Granulocytic Leukemias, Chronic):ti,ab,kw OR (Myeloid Leukemia, Philadelphia Positive):ti,ab,kw (Word variations have been searched) #7 #2 or #3 or #4 or #5 or #6 #8 #1 and #7	
	Imatinib	#1 (Imatinib Mesylate):ti,ab,kw OR (Gleevec):ti,ab,kw OR (Glivec):ti,ab,kw OR (Methanesulfonate, Imatinib):ti,ab,kw OR (Mesylate, Imatinib):ti,ab,kw (Word variations have been searched) #2 (Imatinib Methanesulfonate):ti,ab,kw OR (ST 1571):ti,ab,kw OR (ST1571):ti,ab,kw OR (STI571):ti,ab,kw OR (STI-571):ti,ab,kw (Word variations have been searched) #3 (STI 571):ti,ab,kw OR (CGP57148B):ti,ab,kw OR (CGP 57148):ti,ab,kw OR (CGP57148):ti,ab,kw OR (CGP-57148):ti,ab,kw (Word variations have been searched) #4 #1 or #2 or #3 #5 (Leukemia, Myelogenous, Chronic, BCR-ABL Positive):ti,ab,kw OR (Philadelphia-Positive Myeloid Leukemia):ti,ab,kw OR (Myeloid Leukemias, Chronic):ti,ab,kw OR (Leukemia, Myelogenous, Ph1 Positive):ti,ab,kw OR (Leukemias, Chronic Granulocytic):ti,ab,kw (Word variations have been searched) #6 (Leukemia, Myelogenous, Chronic, BCR-ABL Positive):ti,ab,kw OR (Myelogenous Leukemias, Ph1 Positive):ti,ab,kw OR (Leukemia, Myeloid, Philadelphia Positive):ti,ab,kw OR (Myelocytic Leukemias, Chronic):ti,ab,kw OR (Granulocytic Leukemia, Chronic):ti,ab,kw (Word variations have been searched) #7 (Chronic Myelogenous Leukemia):ti,ab,kw OR (Leukemias, Ph1 Positive Myelogenous):ti,ab,kw OR (Myelogenous Leukemias, Chronic):ti,ab,kw OR (Chronic Granulocytic Leukemia):ti,ab,kw OR (Leukemia, Myelogenous, Ph1 Positive):ti,ab,kw (Word variations have been searched) #8 (Leukemia, Philadelphia-Positive Myeloid):ti,ab,kw OR (Ph1 Positive Myeloid Leukemia):ti,ab,kw OR (Myeloid Leukemia, Chronic):ti,ab,kw OR (Leukemia, Myeloid, Ph1 Positive):ti,ab,kw OR (Leukemia, Granulocytic, Chronic):ti,ab,kw (Word variations have been searched)	329

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		#9 (Leukemia, Ph1 Positive Myeloid):ti,ab,kw OR (Chronic Myelocytic Leukemia):ti,ab,kw OR (Myeloid Leukemias, Ph1 Positive):ti,ab,kw OR (Granulocytic Leukemias, Chronic):ti,ab,kw OR (Myeloid Leukemia, Philadelphia Positive):ti,ab,kw (Word variations have been searched) #10 #5 or #6 or #7 or #8 or #9 #11 #4 and #10	

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eTable 2: Characteristics of included studies and quality assessment.

Authors (year)	Trial register	Trial desig	Country	Treatment arm	Number of	Population	Median age	Male,n (%)	Jadad
Gambacorti-Passerini et al,[23]	NCT00574873	Phase III; RCT	30 countries	bosutinib	250	chronic phase CML	48(19-	149(59.	2
				imatinib 400	252		47(18-89)	135(53.6)	
Cortes et al,[24] 2018	NCT02130557	Phase III;	27 countri	bosutinib	268	chronic phase	53(18-	156(58.	2
				imatinib	268		53(19-	155(57.	
Hjorth-Hansen et al,[25] 2014	NCT00852566	Phase II; RCT	Finland, Norway	dasatinib	22	chronic phase CML	53(29-	7(31.8)	2
				imatinib 400	24		58(38-78)	15(62.5)	
Kantarjian et al,[26]	NCT00481247	Phase III;	28 countri	dasatinib	259	chronic phase	46(18-	144(56)	3
				imatinib	260		49(18-	163(63)	
Radich et al,[27] 2012	NCT00070499	Phase II;	Canada, United	dasatinib	123	chronic phase	47(18-	75(61)	3
				imatinib	123		50(19-	72(58.5)	
Hughes et al,[28] 2014	NCT00760877	Phase III;	6 countri	nilotinib	104	chronic phase	46 (23-	71(68.3)	2
				imatinib	103		52 (19-	65(63.1)	
Saglio et al,[29] 2010	NCT00471497	Phase III;	36 countri	nilotinib	563	chronic phase	47(18-	333(59.	3
				imatinib	283		46(18-	158(56)	
Wang et al,[30] 2015	NCT01275196	Phase III;	China	nilotinib	134	chronic phase	41(18-	91(67.9)	3
				imatinib	133		39(19-	81(60.9)	
Lipton et al,[31] 2016	NCT01650805	Phase III;	25 countri	ponatinib	154	chronic phase	55 (18-	97(63)	3
				imatinib	152		52 (18-	92(61)	

Abbreviation: CML = Chronic myeloid leukemia.

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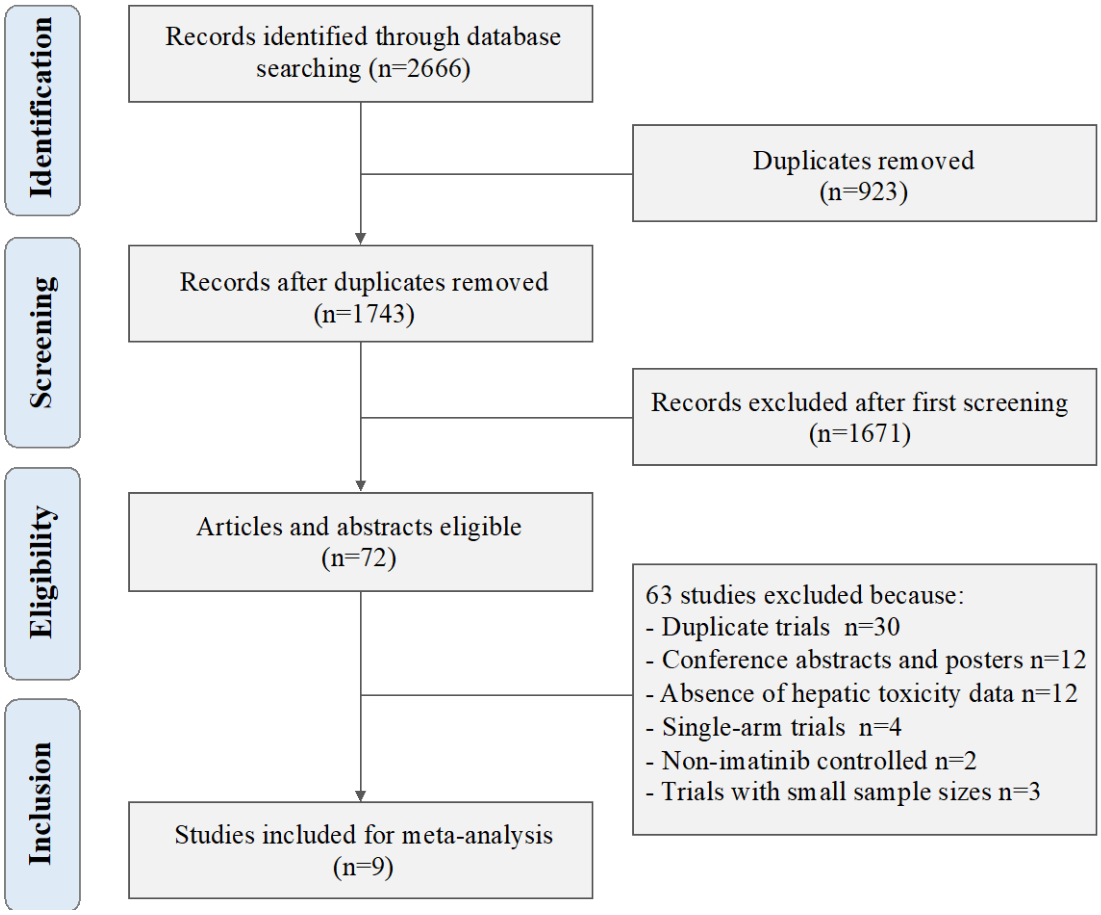
eFigure 1: Flow diagram of study selection according to PRISMA statement.



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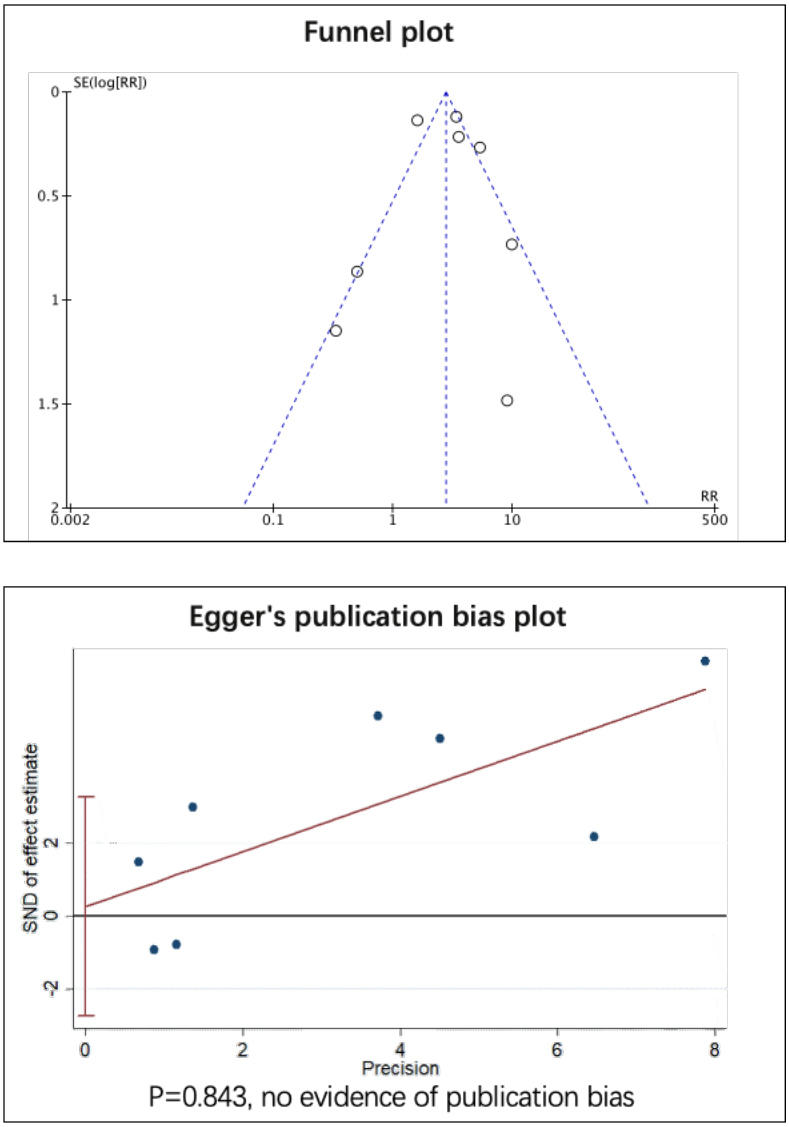
eFigure 2: Evaluation of publications bias.

A. All grades alanine aminotransferase (ALT) elevation



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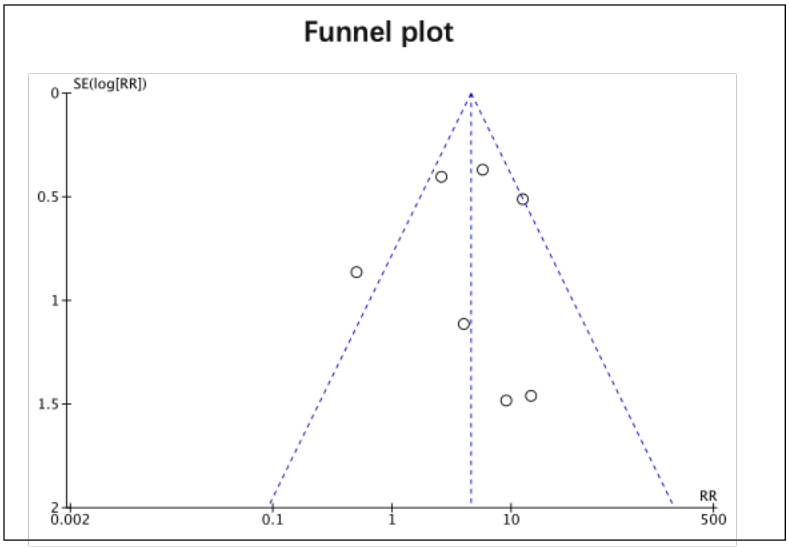
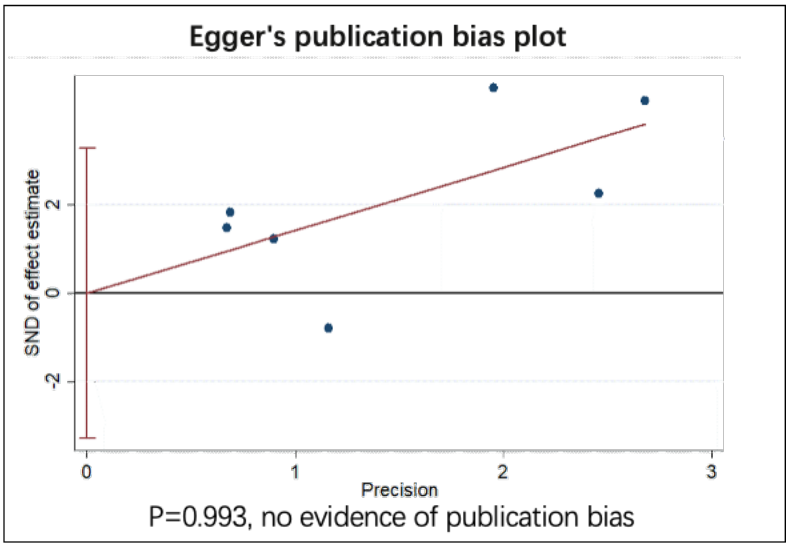
B. Grade 3-4 ALT elevation





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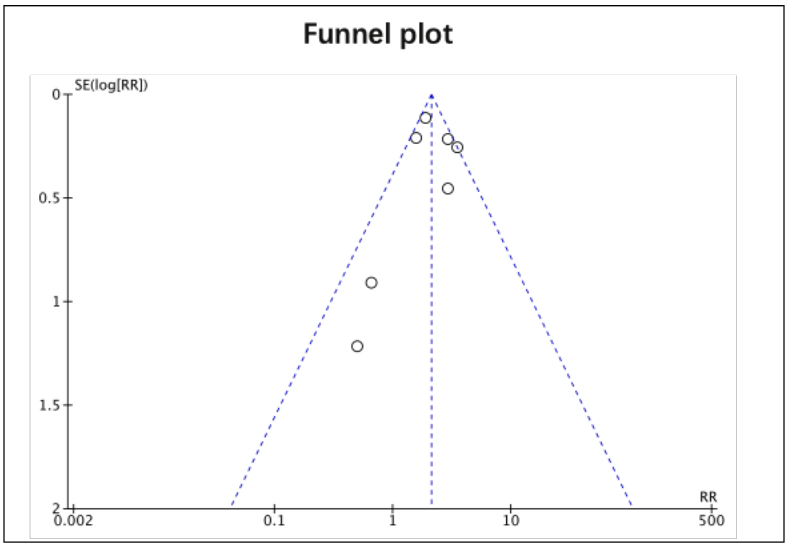
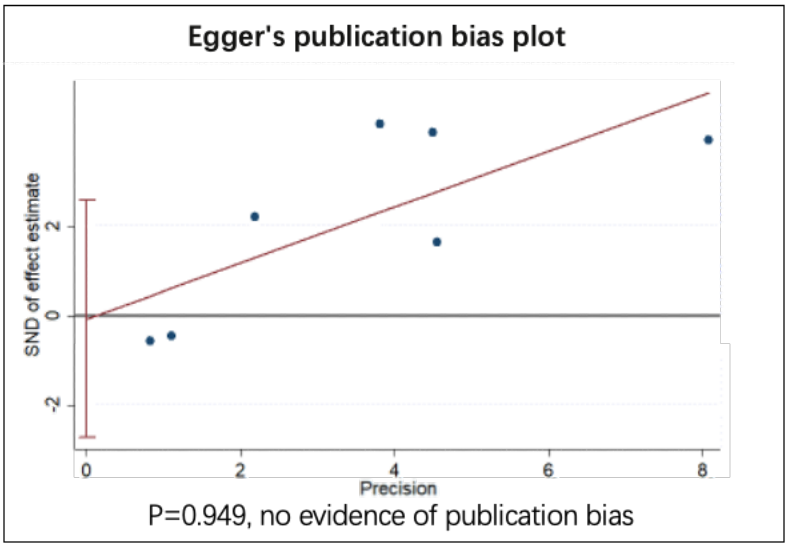
C. All grades AST elevation





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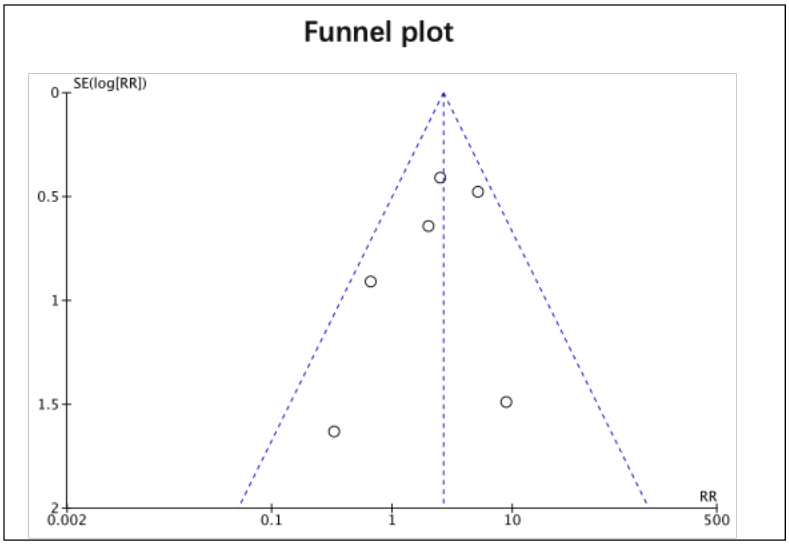
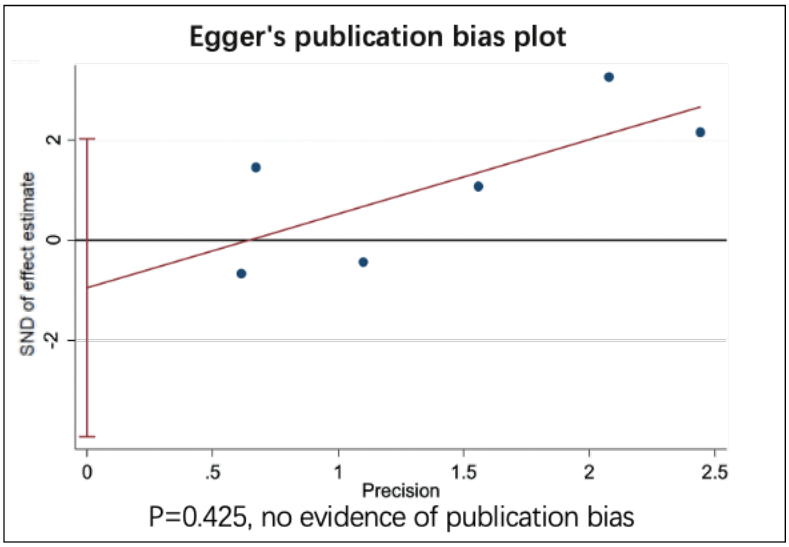
D. Grade 3-4 aspartate aminotransferase (AST) elevation





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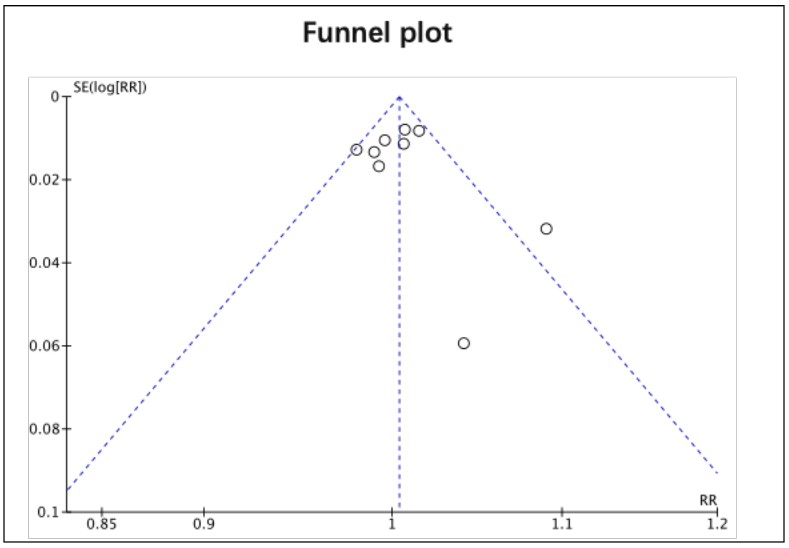
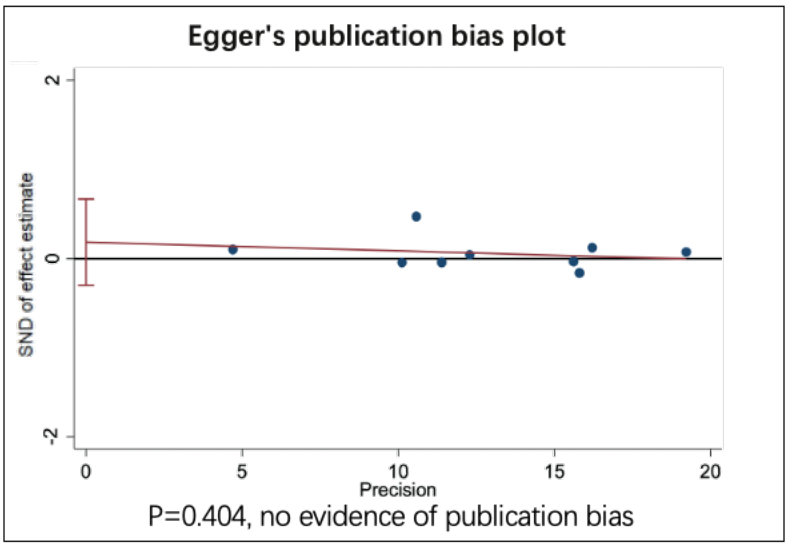
E. Overall survival (OS)





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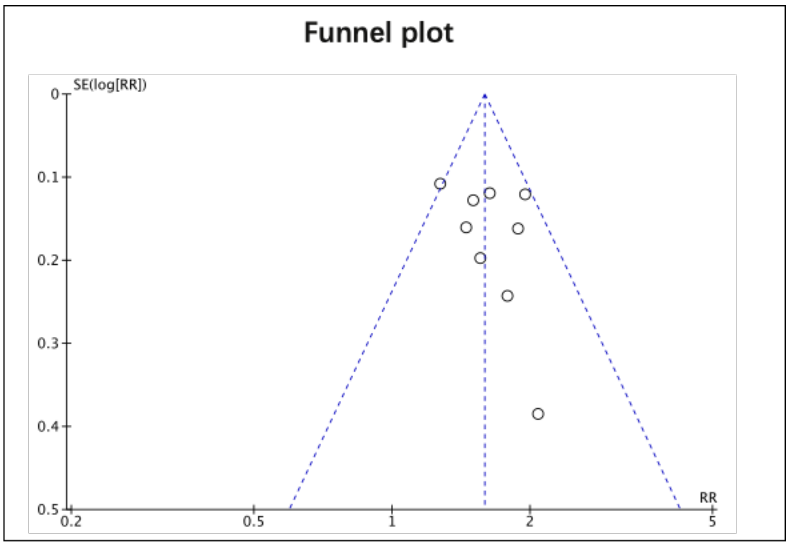
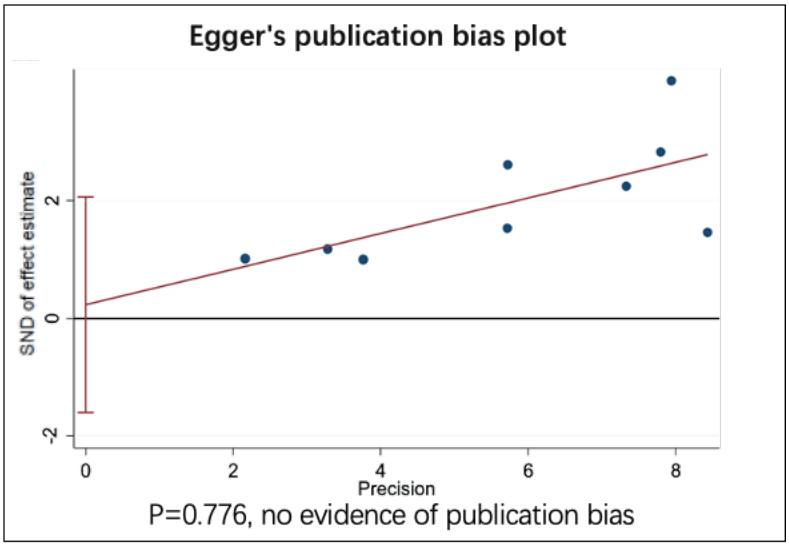
F. Major molecular response (MMR)





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